Excel Assignment 2: Data Cleaning and Transformation

1. Executive Overview & Problem Statement

In real-world data analytics pipelines, unprocessed source data frequently exhibits null values, casing inconsistencies, unstandardized categorizations, duplicate entries, and concatenated fields that impede business intelligence and statistical modeling. The objective of this assignment is to execute a rigorous data cleaning protocol on the 34-row Product Dataset, transforming it into an analysis-ready asset adhering to strict enterprise standards.

2. Task 1: Handling Missing Values

Identification: The dataset contains 3 records with missing prices and 4 records with missing categories.
Category Imputation Strategy: Category attributes were resolved by cross-referencing identical product records in the catalog. *Sneakers* was mapped to 'Fashion' (matching Nike Sneakers), *Coffee Maker* to 'Kitchen' (matching Keurig Coffee Maker), *Fitness Tracker* to 'Electronics' (matching Garmin Fitness Tracker), and *Backpack (North Face)* to 'Outdoor'.
Price Imputation Strategy: Missing prices were imputed using direct product-type benchmark pricing: Sony Headphones was set to $250.00 (matching Bose Headphones), Ray-Ban Sunglasses was set to $150.00 (matching Oakley Sunglasses), and Coleman Camping Tent was set to $130.00 (matching the Outdoor category benchmark).

Step 1: Missing Value Identification & Imputation in Excel

Product ID	Product Name	Brand Name	Price ($)	Qty	Category
22-MAY-US	Backpack	North Face	$70.00	20	[MISSING] -> Outdoor (Imputed)
07-JUN-UK	Headphones	Sony	[MISSING] -> $250.00 (Imputed)	45	Electronics
25-NOV-AU	Sneakers	Adidas	$90.00	40	[MISSING] -> Fashion (Imputed)
08-DEC-DE	Coffee Maker	Nespresso	$120.00	35	[MISSING] -> Kitchen (Imputed)
14-AUG-US	Camping Tent	Coleman	[MISSING] -> $130.00 (Imputed)	10	Outdoor
19-JUL-BR	Fitness Tracker	Xiaomi	$150.00	30	[MISSING] -> Electronics (Imputed)
03-JUN-CA	Sunglasses	Ray-Ban	[MISSING] -> $150.00 (Imputed)	25	Fashion

Strategy: Missing categories filled by product lookup; missing prices imputed via product/category benchmark values.

3. Task 2: Correcting Inconsistent Data

Text Inconsistency: In the 'Product Name' column, multiple records suffered from lowercase styling ('laptop', 'smartphone', 'headphones'). This was standardized to Title Case using Excel's PROPER() function and Find & Replace.
Category Typos: The Category column had 5 entries recorded as 'Electroni'. A targeted Find and Replace operation (Find: 'Electroni', Replace: 'Electronics') was applied across the dataset.

Step 2: Find & Replace - Standardizing Text & Fixing Typos

Product ID	Product Name (Standardized)	Brand	Price	Qty	Category (Typo Fix)
28-JAN-US	laptop -> Laptop	Dell	$1,000.00	30	Electronics (Correct)
11-APR-US	smartphone -> Smartphone	Samsung	$900.00	25	Electronics (Correct)
07-JUN-UK	Headphones	Sony	$250.00	45	Electroni -> Electronics (Fixed)
18-FEB-CA	Smartwatch	Fitbit	$150.00	15	Electroni -> Electronics (Fixed)
16-APR-ES	headphones -> Headphones	Bose	$250.00	20	Electroni -> Electronics (Fixed)
27-JAN-IT	laptop -> Laptop	Asus	$980.00	10	Electronics (Correct)
20-AUG-CN	Smartwatch	Huawei	$160.00	15	Electroni -> Electronics (Fixed)

Operations: PROPER() applied to casing; Find & Replace replaced 'Electroni' with 'Electronics'.

4. Task 3: Removing Duplicate Records

Audit & Detection: A multi-column uniqueness scan revealed 3 exact duplicate rows: HP Laptop (17-JUN-IN), Bose Headphones (16-APR-ES), and Samsonite Laptop Bag (21-AUG-CA).
Excel Procedure: Data tab > Data Tools > *Remove Duplicates* (with all columns selected). This cleanly trimmed the dataset from 34 rows to 31 unique records.

Step 3: Duplicate Record Detection and Removal (3 Duplicates Removed)

Product ID	Brand	Product Name	Price	Qty	Category	Deduplication Status
17-JUN-IN	HP	Laptop	$950.00	25	Electronics	Retained (Original)
17-JUN-IN	HP	Laptop	$950.00	25	Electronics	DELETED (Exact Duplicate)
16-APR-ES	Bose	Headphones	$250.00	20	Electronics	Retained (Original)
16-APR-ES	Bose	Headphones	$250.00	20	Electronics	DELETED (Exact Duplicate)
21-AUG-CA	Samsonite	Laptop Bag	$50.00	35	Accessories	Retained (Original)
21-AUG-CA	Samsonite	Laptop Bag	$50.00	35	Accessories	DELETED (Exact Duplicate)

Excel Feature: Data Tab -> Remove Duplicates across all 6 columns. Total rows reduced from 34 to 31.

5. Task 4: Splitting and Merging Data

Column Splitting: The composite 'Product ID' (e.g., '28-JAN-US') was parsed via hyphen delimiter '-' into two distinct fields: **Manufacturing Date** (parsed into a full date structure) and **Country Code** (ISO 2-letter alpha code).
Column Merging: 'Brand Name' and 'Product Name' were combined using =E3 & " " & F3 to produce a unified **Product Brand** column (e.g., 'Dell Laptop', 'Nike Sneakers').

Step 4: Splitting Product ID & Merging Product Brand

Original ID	Manufacturing Date	Country Code	Product Brand (Merged)	Brand Name	Product Name
28-JAN-US	28-01-2024	US	Dell Laptop	Dell	Laptop
15-FEB-US	15-02-2024	US	Nike Sneakers	Nike	Sneakers
03-MAR-US	03-03-2024	US	Keurig Coffee Maker	Keurig	Coffee Maker
07-JUN-UK	07-06-2024	UK	Sony Headphones	Sony	Headphones
18-FEB-CA	18-02-2024	CA	Fitbit Smartwatch	Fitbit	Smartwatch
21-AUG-CA	21-08-2024	CA	Samsonite Laptop Bag	Samsonite	Laptop Bag

Transformations: Date & Country split via Text-to-Columns / Delimiter '-'; Brand and Name merged with '&'.

6. Tasks 5 & 6: Number Formatting & Conditional Formatting

Number Formatting: The Price column was formatted as Currency ($#,##0.00). The Manufacturing Date was formatted to the strict standard DD-MM-YYYY.

Conditional Formatting: Applied a custom rule on Category: Cell Value = "Electronics" with soft-green background fill (#E2EFDA) and dark green text. In addition, applied gradient Data Bars across the Price column to visually illustrate product price tiers.

Step 5 & 6: Currency Formatting & Conditional Formatting Rules

Product Brand	Category (Highlight Rule)	Price (Currency)	Price Visual Data Bar
Dell Laptop	Electronics	$1,000.00	(Max)
HP Laptop	Electronics	$950.00	
Samsung Smartphone	Electronics	$900.00	
Nikon Camera	Electronics	$700.00	
Sony Headphones	Electronics	$250.00	
Keurig Coffee Maker	Kitchen	$130.00	
Nike Sneakers	Fashion	$80.00	
Zara Dress	Fashion	$60.00	
Samsonite Laptop Bag	Accessories	$50.00	

Rules: Green fill on Category='Electronics'; Currency format '$#,##0.00'; Gradient data bars on Price.